

RentEase — 30 Day Development Roadmap

Week 1 — Foundation & Backend Architecture

Day 1 — Project Initialization & MERN Setup

- Initialize MERN project
- Setup React + Vite frontend
- Setup Express backend
- Configure Tailwind CSS
- Create routing structure
- Build landing page foundation
- Connect frontend and backend

Day 2 — Backend Foundation & MongoDB

- Configure MongoDB Atlas
- Setup environment variables
- Create middleware architecture
- Add centralized error handling
- Setup backend API structure
- Prepare authentication-ready backend

Day 3 — Database Schema Architecture

- Create User schema
- Create Property schema
- Create Booking schema
- Create Review schema
- Add model relationships
- Optimize schemas and indexing

Day 4 — Authentication System

- Firebase Authentication
- JWT backend authentication
- Login/Register flows

- Google sign-in
- Protected routes
- Role-based access control

Day 5 — Property Management System

- Property CRUD APIs
- Property listing UI
- Property details page
- Owner property management
- Property filtering/search foundation

Day 6 — Booking & Reservation System

- Booking APIs
- Booking validation
- Date conflict prevention
- My Bookings dashboard
- Owner booking management
- Booking status system

Day 7 — Razorpay Payment Integration

- Razorpay integration
- Payment verification
- Advance booking payments
- Payment success/failure pages
- Payment dashboard
- Transaction handling

Week 2 — Advanced Core Features

Day 8 — Image Upload System

- Cloudinary integration
- Multi-image uploads

- Property image gallery
- Image preview/removal
- Backend upload handling

Day 9 — Google Maps Integration

- Google Maps API setup
- Property location pins
- Interactive map view
- Geolocation support
- Nearby property exploration

Day 10 — Advanced Search & Filtering

- Search by city/location
- Price range filters
- Amenities filters
- Property type filters
- Sorting system
- Search optimization

Day 11 — Reviews & Ratings System

- Property reviews
- Star ratings
- Review CRUD
- Average rating calculation
- Review display UI

Day 12 — Wishlist & Favorites

- Save properties
- Favorite management
- Wishlist dashboard
- Quick save/remove actions

Day 13 — Notifications System

- Toast notifications
- Booking notifications
- Payment notifications
- Owner alerts
- Real-time status messages

Day 14 — Responsive UI Polish

- Mobile responsiveness
 - Tablet optimization
 - Navbar improvements
 - UI consistency
 - Accessibility improvements
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Week 3 — Dashboards & Advanced Features

Day 15 — Tenant Dashboard Enhancement

- Booking analytics
- Payment summaries
- Recent activity
- Profile management
- Saved properties

Day 16 — Owner Dashboard Enhancement

- Property analytics
- Revenue overview
- Booking insights
- Occupancy stats
- Property management improvements

Day 17 — Admin Dashboard

- User management
- Property moderation

- Booking monitoring
- Payment oversight
- Platform statistics

Day 18 — Profile Management System

- Edit profile
- Avatar upload
- Password update
- Account settings
- Contact management

Day 19 — Chat/Messaging Foundation

- User-owner messaging
- Conversation UI
- Message APIs
- Chat layout
- Realtime preparation

Day 20 — Real-Time Chat Integration

- Socket.IO integration
- Live messaging
- Online indicators
- Instant updates
- Notification integration

Day 21 — Email Notification System

- Booking confirmation emails
 - Payment receipts
 - Welcome emails
 - Owner alerts
 - Nodemailer integration
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Week 4 — Optimization, Security & Deployment

Day 22 — Security Improvements

- Rate limiting
- Input sanitization
- Helmet security
- XSS protection
- Secure headers
- API protection

Day 23 — Performance Optimization

- Lazy loading
- API optimization
- Pagination
- Caching improvements
- Query optimization

Day 24 — Advanced Booking Features

- Booking cancellation policy
- Refund logic foundation
- Booking calendar
- Availability management
- Reservation improvements

Day 25 — Analytics & Insights

- Dashboard charts
- Booking analytics
- Revenue charts
- Admin insights
- Reporting foundation

Day 26 — Testing & Bug Fixing

- Frontend testing

- Backend API testing
- Edge-case validation
- Payment flow testing
- UI bug fixes

Day 27 — Deployment Preparation

- Production environment setup
- Environment cleanup
- Build optimization
- Deployment configs
- README improvement

Day 28 — Deployment

- Deploy frontend to Vercel
- Deploy backend to Render
- Configure production database
- Configure environment variables
- Production testing

Day 29 — Final Production Improvements

- SEO improvements
- Error pages
- Loading skeletons
- Final UI polish
- Production fixes

Day 30 — Portfolio & Resume Ready

- Professional README
- Demo video
- Screenshots
- Resume project entry
- Portfolio integration

- Final GitHub cleanup
- Launch-ready review